

Matthew Oj

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SKILLS

Languages: Python, SQL, JavaScript/TypeScript, PHP, HTML/CSS

Frameworks & Cloud: FastAPI, Next.js, React, Astro, Laravel, Docker, GitHub Actions, AWS, GCP, Vercel, Netlify, CI/CD

Databases & Data Tools: PostgreSQL, SQLite, MongoDB, BigQuery, Snowflake, dbt, Apache Airflow, Snowpipe, DuckDB, XGBoost, SHAP, Vertex AI
Generative Models

EXPERIENCE

Vafo Digital, Senior Data Engineer (Backend Infrastructure) Apr 2026 – present

- Led migration of fragmented backend orchestration (Snowflake procedures, PHP services, serverless functions) to a unified self-hosted platform on AWS EC2 using Docker, provisioning isolated staging/production environments with an Aurora RDS metadata store and IAM-scoped access, consolidating scheduling for 100+ services
- Architected a reporting service (FastAPI, AWS Lambda, Mangum) automating scheduled delivery across 10 client accounts, eliminating manual ad hoc requests
- Built a shared Python library packaged for reuse, standardising error handling and logging across 30+ services and 4 integrations (Klaviyo, DynamoDB, SFTP, S3)
- Developed a self-hosted dbt documentation portal, generating manifest artefacts via dbt docs, publishing to S3, and serving through AWS Amplify with a custom HTML, CSS and JavaScript front end, giving stakeholders searchable, always-current data lineage

Vafo Digital, Data Engineer Apr 2025 – Apr 2026

- Collaborated with the Data Science Lead to productionise 4 pipelines (Marketing Mix Modelling, personalisation, churn) using dbt feature engineering, XGBoost/SHAP training, and Snowflake Model Registry for versioning
- Built and hosted 3 client-facing analytics platforms (Python, Streamlit) on Snowflake, with high-performance querying against event tables exceeding 200M+ rows to deliver real-time insights to non-technical stakeholders
- Developed Python AWS Lambda services syncing backend data to external systems (HubSpot, Klaviyo), supporting 250K+ users

Maison Privé Concierge Limited, Software Engineer (Freelance) | maisonprive.london Apr 2026 – Jun 2026

- Built and deployed a luxury concierge marketing site (Astro, TypeScript) on Vercel, with SSR API routes handling enquiries across 9 services in 3 categories
- Engineered a dynamic enquiry form with per-service field schemas, validation, rate limiting, and spam filtering, blocking 95%+ of bot submissions
- Built a dual-sink pipeline routing enquiry to Mailgun and Airtable in real time

EssenceMediacom, Data Software Engineer Nov 2023 – Apr 2025

- Built a Cloud Run-hosted web app for table schema generation, cutting pipeline build time by 30% via no-code tool integration
- Improved Apache Airflow DAG deployment speed by 40% through CI/CD pipeline optimization with GitHub Actions
- Developed a REST API-driven alert system with Cloud Run to notify teams of Adverity DataStream failures, boosting incident response time
- Developed a secure web application for the Data Strategy team, enabling OAuth-based access for users across 5+ agencies

PROJECTS

Finance Tracker | FastAPI, dbt, PostgreSQL, Next.js, Docker | [GitHub](#)

- Designed a production-style data stack ingesting raw financial data through dbt models into aggregated PostgreSQL marts, served via a JWT-secured FastAPI and visualised in a Next.js dashboard, all containerised with Docker Compose
- Wrote a pytest suite covering auth, token validation, all endpoints, and pagination against a fully mocked database - 13/13 tests passing.

Event-Driven Notification Service | Python, FastAPI, Kafka, aiokafka, PostgreSQL, Docker | [GitHub](#)

- Built an async event-driven notification service processing order events via Kafka, with idempotent PostgreSQL upserts to safely handle at-least-once delivery and message redelivery
- Implemented retry logic with a dead-letter queue for failed messages, exposing order and notification status through a FastAPI service layer

ACTIVITIES

WPP & Google Cloud Hackathon (3rd Place) | Python, Cloud Run, Vertex AI, Vertex AI Generative Models, Streamlit Sep 2024

- Designed and deployed a Cloud Run service pulling real-time data from public BigQuery tables to dynamically generate prompts for Vertex AI, rendering AI-generated advertising creatives through a Streamlit frontend, placed 3rd of all competing teams

Global Game Jam Hackathon - ML & Game Developer | Python, OpenCV, C#, Unity Jan 2024

- Built a Python/OpenCV ML script using camera input to detect laughs, aligned with the hackathon's humour theme
- Contributed Unity/C# game objects and animations to the game prototype

Engineers Without Borders - Project Lead Engineer | MATLAB, SolidWorks Sep 2020 – May 2021

- Engineered a MATLAB script to optimize thermal efficiency in material selection – increasing solar cooker design efficiency by 20%.
- Led a team of 5 engineers to design a solar cooker prototype with phase-changing materials using SolidWorks

EDUCATION

University of Greenwich Sep 2020 – May 2023

Bachelor of Engineering, Civil Engineering, 2.1 Upper Second-Class Honours

London, UK

Dissertation (First Class): Assessing the vulnerability of catchments during flash flood events in South England using Machine Learning methods